

Circuit Analysis (Lab11)

Experiment 10: Verification of Norton's Theorems

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Date: 15th May, 2026..... Grade and Signature:

CLO2 CLO3	Use with identification the circuit components, breadboards, multimeter, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)				
	Total marks (10)			

Objective

The objective of this lab is verify Norton's theorem and to find the full load current.

Apparatus Required

Sr. no	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	1k, 330 ohms	1,3
7	Potentiometer	Standard	1

Description

Statement:

Any linear circuit can be simplified to an equivalent circuit consisting of a single current source and parallel resistance that is connected to a load.

Precautions:

1. Voltage control knob of RPS should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position

Procedure for Norton's Theorem:

1. Connections are given as per circuit diagram.
2. Set a particular value in RPS and note down the ammeter readings in the original circuit.

To Find I_N :

3. Remove the load resistance and short circuit the terminals.
4. For the same RPS voltage note down the ammeter readings.

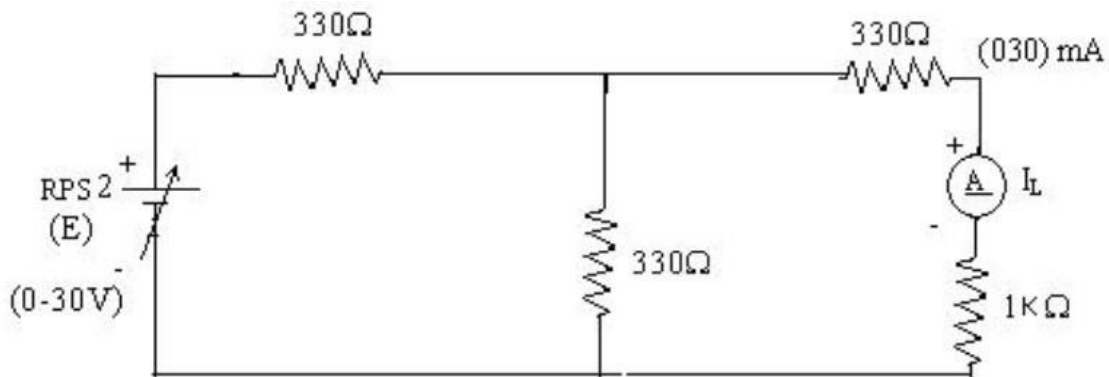
To Find R_N :

5. Remove RPS and short circuit the terminal and remove the load and note down the resistance across the two terminals.

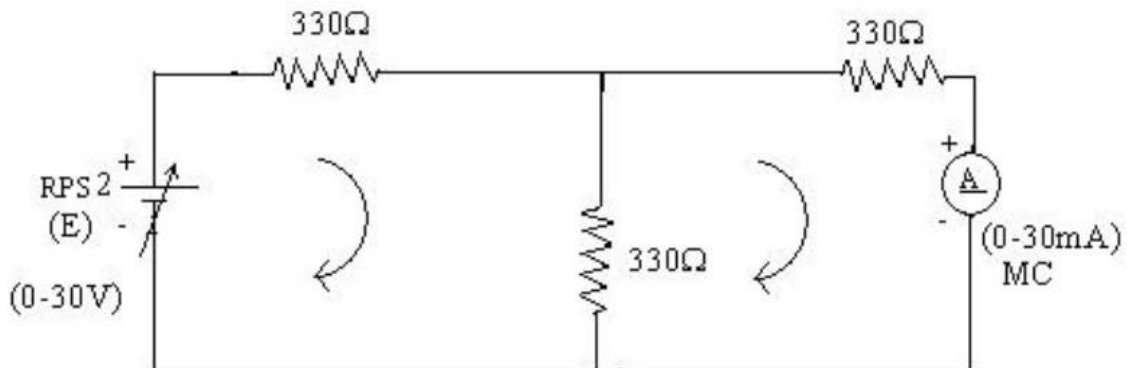
Equivalent Circuit:

6. Set I_N and R_N and note down the ammeter readings.
7. Verify Norton's theorem.

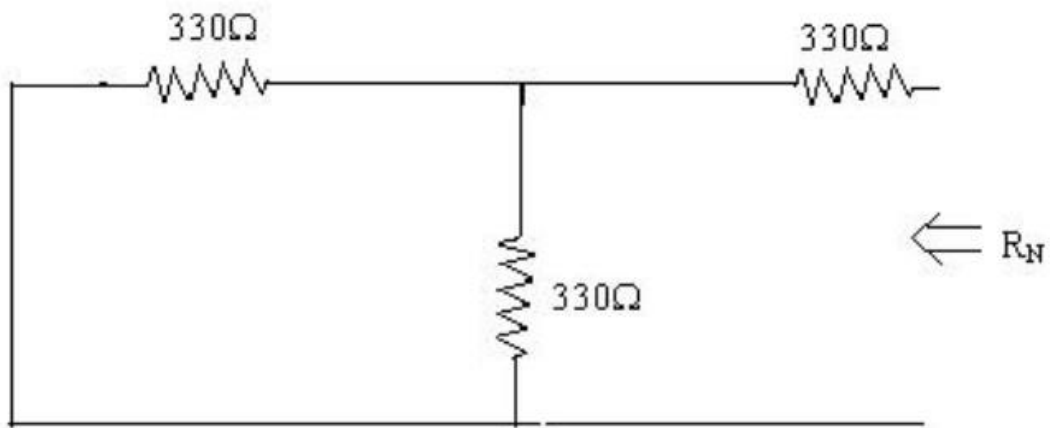
To find load current in circuit 1:



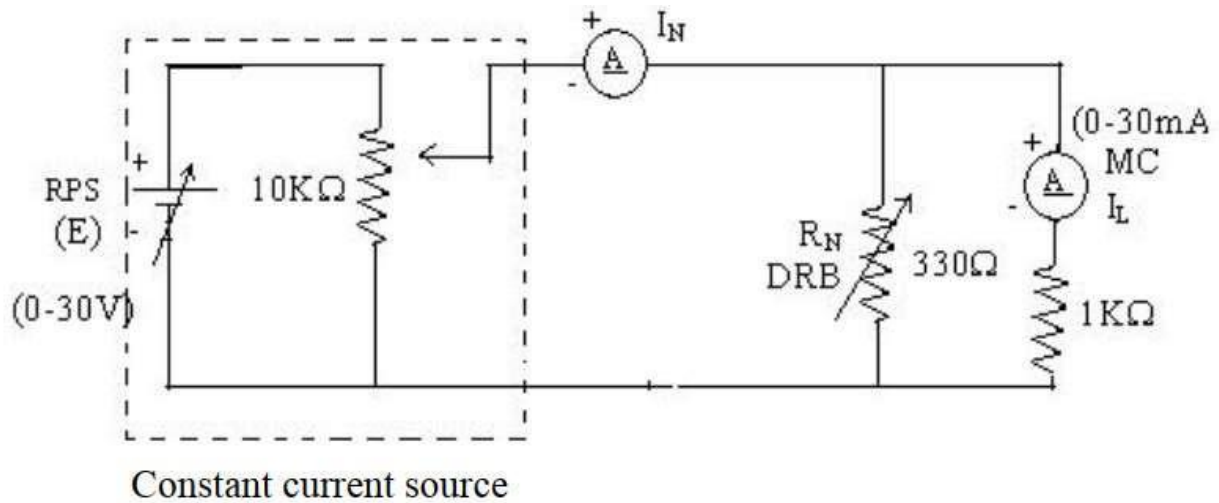
To find I_N



To find R_N



Norton's equivalent circuit



Norton's Practical Values

	E (volts)	I_N (mA)	R_N (ohms)	I_L (mA)	
				Circuit - I	Equivalent Circuit
Practical Values	10	10.1	495	3.34	3.34

Post Lab Questions

1. What is Norton's Theorem?

Norton's Theorem states that any linear circuit consisting of voltage sources, current sources and resistors can be simplified into an equivalent circuit. This simplified circuit consists of just two components, a single constant current source (Norton Current) connected in parallel with a single equivalent resistor (Norton Resistance).

2. What are the applications of Norton's Theorem?

The main application is simplifying complex electrical networks to make them easier to analyze. It is especially useful when you have a circuit where only one specific load resistor frequently changes its value. Instead of recalculating the entire complex circuit every time the load changes you only have to calculate the current flow using the simple two component Norton equivalent. It is also heavily used to calculate the maximum power transfer in a system and short circuit analysis.

3. What are the theoretical and practical limitations of Norton's Theorem?

Theoretical Limitations: The theorem strictly applies only to linear and bilateral circuits. It completely fails if the circuit contains non-linear components (like diodes or transistors) where resistance changes with voltage or current. It also cannot be applied to circuits with magnetic coupling (like certain transformers) between the load and the rest of the network.

Practical Limitations: In the real world a perfect "ideal" constant current source does not exist. Additionally measuring the Norton current practically requires creating a direct short circuit across the output terminals. While this is easy in simulation, doing this to a real high-power physical supply in the lab can cause dangerous current spikes, excessive heat or permanent damage to the equipment.